BUSINESS PROPOSAL

"SuryaSheetal" – Solar-Powered Micro Cold Storage Network for Indian Farmers

"Store Today. Sell Better Tomorrow."

1. Executive Summary

India is one of the world's largest agricultural producers, yet farmers often face significant losses because fruits, vegetables, dairy products and other perishables cannot be stored properly after harvesting.

A major reason is that conventional cold-storage facilities are often expensive, centralized and located far from farms. Small and marginal farmers may not have enough produce to justify renting an entire storage facility, while unreliable electricity in rural areas further increases the difficulty.

SuryaSheetal proposes a network of small, solar-powered, temperature-controlled cold-storage units located close to farming communities. Farmers can rent storage space on a pay-per-crate/per-day basis rather than investing in their own cold-storage infrastructure.

The units would use:

- Solar photovoltaic panels
- Battery/thermal energy storage
- Efficient refrigeration
- IoT temperature and humidity sensors
- Mobile/WhatsApp-based booking
- Digital inventory tracking
- Local operators/FPOs for management

The business earns revenue through storage fees, transportation, subscription plans and B2B partnerships. The model simultaneously addresses: **Food wastage + farmer income + renewable energy + rural employment + supply-chain efficiency.**

2. Problem Statement

Farmers frequently have to sell perishable produce immediately after harvesting because they lack affordable nearby storage.

This creates a situation where:

Harvest → No nearby storage → Forced immediate sale → Oversupply → Lower prices → Spoilage → Farmer income loss

For example, during peak harvesting seasons, farmers may face a sudden fall in prices because large quantities of the same crop enter the market simultaneously. A farmer may therefore have two choices:

1. Sell immediately at a low price, or
2. Keep the produce without proper refrigeration and risk spoilage.

This is particularly problematic for: Tomatoes, Potatoes, Onions, Fruits, Leafy vegetables, Milk and dairy products, Flowers, and other temperature-sensitive agricultural products.

3. Why Existing Cold Storage Is Not Enough

Traditional cold storage has several limitations for small farmers:

- **1. High capital cost:** Building a large cold-storage facility requires substantial investment.
- **2. Distance:** Large cold stores are generally concentrated around major agricultural and commercial centers rather than every farming cluster.
- **3. Small quantities:** A small farmer may only have a few crates of produce and may not find conventional storage economically convenient.
- **4. Electricity dependency:** Conventional refrigeration requires reliable electricity, which can be a challenge in some rural areas.
- **5. Lack of digital access:** Farmers may not know available storage capacity, storage price, expected storage duration, or nearby facilities.

4. Proposed Solution: SuryaSheetal

SuryaSheetal will establish small modular cold-storage units directly in farming clusters. Instead of creating one huge cold-storage facility, we create a network of smaller units.

Farm → SuryaSheetal → Temporary Storage → Better Market Timing → Buyer

Each unit can serve approximately 20–50 farmers, depending on capacity and local demand. Farmers don't have to purchase the infrastructure. They simply pay for the storage they use.

5. How the System Works

- **Step 1 – Farmer Harvests Produce:** A farmer harvests tomatoes, fruits, vegetables or another eligible crop.
- **Step 2 – Farmer Books Storage:** The farmer books storage through a Mobile application, WhatsApp, SMS, or Local SuryaSheetal operator.
- **Step 3 – Produce Is Collected:** The farmer can either bring produce to the unit, or request pickup through the SuryaSheetal network.
- **Step 4 – Quality Check:** Produce is checked for quantity, crop type, quality, and existing damage.

- **Step 5 – Storage:** Produce is placed inside the temperature-controlled chamber. The appropriate temperature is selected according to the crop.
- **Step 6 – Monitoring:** IoT sensors continuously monitor Temperature, Humidity, Door opening, and Power status.
- **Step 7 – Notification:** If temperature goes outside the safe range, the operator receives an alert.
- **Step 8 – Farmer Decides When to Sell:** The farmer can now wait for better market prices, a suitable buyer, transportation availability, or reduced market oversupply.

6. Technology

The technology is highly effective and optimized for rural deployment:

> **Hardware Flow:**
> Solar Panels → Charge Controller → Battery / Thermal Storage → Efficient Refrigeration System → Insulated Cold Chamber
>
> **Monitoring Flow:**
> Temperature + Humidity Sensors → IoT Controller → Cloud Dashboard → Farmer/ Operator Alerts

7. Smart Features

A. Smart Temperature Control

Different crops require different storage conditions. The system automatically maintains the required range.

B. IoT Monitoring

The operator can see metrics like *Temperature: 8°C, Humidity: 85%, Capacity: 72%, Solar Battery: 81%* from a mobile dashboard.

C. Automated Alerts

If the temperature rises unexpectedly, the operator receives an immediate notification (e.g., ⚠️ *Temperature Alert: Unit #12*).

D. Digital Inventory

Every stored batch receives a digital ID (e.g., Farmer: Ramesh Kumar | Crop: Tomato | Quantity: 12 crates | Storage Date: 15 Sept).

E. Market-Price Integration

A future version can display nearby mandi/market prices. The platform presents market information as decision support.

Market	Tomato Price
Local Mandi	₹12/kg
City Mandi	₹18/kg
Nearby Wholesale Market	₹20/kg

8. Renewable Energy Model

The biggest differentiator is the use of solar energy.

- **During the day:** Sunlight → Solar Panels → Electricity. Used for refrigeration, sensors, lighting, and control systems. Excess energy charges batteries.
- **At night or low sunlight:** Battery/Thermal Storage → Refrigeration. This reduces dependence on diesel generators.

India already has a strong policy ecosystem supporting decentralized solar in agriculture, such as the PM-KUSUM scheme which supports solarization for farmers and FPOs.

9. Business Model

SuryaSheetal is designed as a social-impact business with multiple revenue streams:

- **Revenue Stream 1 – Storage Fees:** Farmers pay ₹ per crate per day. *(e.g., 10 crates × 5 days × ₹4 = ₹200)*
- **Revenue Stream 2 – Subscription Plans:** Monthly/seasonal plans (e.g., Basic Farmer Plan ₹499/month for credits, inventory, alerts).

- **Revenue Stream 3 – Transportation:** Fees for Farm → Cold Storage and Cold Storage → Market transit.
- **Revenue Stream 4 – B2B Storage:** Restaurants, retailers, and wholesalers renting larger capacity.
- **Revenue Stream 5 – Institutional Partnerships:** FPOs, Cooperatives, CSR initiatives, NGOs.

10. Target Customers

Primary Customers: Small and marginal farmers (Fruits, Vegetables, Flowers, Dairy).

Secondary Customers: Farmer Producer Organizations (FPOs), Cooperatives, Local traders, Retailers, Food processors, Restaurants, Wholesalers.

11. Unique Selling Proposition

"A cold store within the farmer's reach, without requiring the farmer to own one."

Traditional: Large + centralized + electricity-dependent + difficult for small quantities.
SuryaSheetal: Small + decentralized + solar-powered + pay-per-use + digitally monitored.

12. Why Farmers Would Use It

Suppose a farmer harvests produce worth ₹20,000 but because of immediate selling and spoilage, realizes only ₹14,000. If affordable storage and better timing help reduce losses and improve selling price, the storage fee becomes economically worthwhile. We sell: **Reduced wastage + flexibility in when to sell + better access to markets.**

13. Sustainability

- **Environmental:** Solar energy reduces fossil-fuel dependence; reduces wasted food and associated resources.
- **Economic:** Recurring revenue from storage and transport means it does not depend on donations.

- **Social:** Creates rural employment (operators, technicians, delivery staff).

14. Feasibility

Solar-powered agricultural cold storage is already being demonstrated in India (e.g., ICAR's Pusa Farm SunFridge). Our innovation is primarily in the **business model + decentralized deployment + digital platform + farmer-friendly pricing + market linkage.**

15–17. Expansion Strategy

Phase 1 – Pilot: Select one agricultural cluster. Install 5–10 micro units serving ~50 villages. Measure utilization, spoilage reduction, and operational cost.

Phase 2 – District Expansion: Scale to 50–100 units. Partner with FPOs, local governments, and banks.

Phase 3 – State-Level Network: Develop a digital cold-chain network where farmers can search for nearby availability and pricing via app.

18. Competitive Advantage

Feature	Traditional Cold Storage	SuryaSheetal
Location	Centralized	Near farms
Energy	Grid/diesel	Solar + backup
Minimum quantity	Often higher	Small quantities
Pricing	Rental-based	Pay-per-use
Booking & Monitoring	Manual, Limited	Digital + IoT + local operator

19. Risk Analysis

- **High Initial Cost:** Use leasing, bank financing, FPO ownership, and CSR partnerships.

- **Low Utilization:** Locate units based on crop production data and demand forecasting.
- **Solar Availability:** Use battery backup, thermal storage, and grid backup where available.
- **Farmer Adoption:** Use local FPOs and village-level operators to build trust.
- **Crop-Specific Requirements:** Start with selected crops and clearly define storage conditions.

20–21. Impact & UN SDGs

Contributes to reduced post-harvest losses, local rural employment, and solar energy usage.

Supports SDGs: 2 (Zero Hunger), 7 (Affordable & Clean Energy), 8 (Decent Work & Economic Growth), 12 (Responsible Consumption & Production), 13 (Climate Action).

22. SWOT Analysis

Strengths	Weaknesses
• Solar-powered • Modular & Near farmers • Pay-per-use & IoT monitoring	• Initial investment required • Requires technical maintenance • Dependence on high utilization
Opportunities	**Threats**
• Huge agricultural market • Government support (KUSUM) • FPO partnerships & Rural digitalization	• Existing cold-storage businesses • Extreme weather & Equipment failure • Market price fluctuations

23. Estimated Pilot Economics

Assume: Storage capacity of 5–10 tonnes, 60% average utilization, effective storage charge of ₹3–₹5 per crate/day plus transport revenue. The core principle for profitability is: **High**

utilization + low energy cost + recurring storage demand = sustainable unit economics.

24–25. Roadmap & Future Expansion

- **Year 1:** Pilot in 1 cluster, gather data, build booking system.
- **Year 2–3:** District & Multi-district expansion (25–100+ units), add B2B partners.
- **Year 4–5 & Future:** State-level network, AI Demand Prediction, Dynamic Pricing, Blockchain Traceability, and Carbon Tracking.

26. Why This Business Can Work in India

India possesses the exact ecosystem needed: A large agricultural sector, increasing solar adoption, established FPOs, wide digital payment and smartphone penetration, and government programs like PM-KUSUM.

27. Final Business Proposition

Solar Energy + **Micro Cold Storage** + **IoT Monitoring** + **Pay-per-use Model** =

Less Food Waste + Better Farmer Income + Sustainable Rural Business

28. Conclusion

India does not only need to produce more food; it also needs to protect the food that has already been produced. SuryaSheetal addresses this problem by bringing affordable cold-chain infrastructure closer to farmers instead of expecting farmers to transport their produce to distant facilities.

The solution is: Innovative, Sustainable, Feasible, Applicable, Profitable, and Socially Impactful.